Optify

Moisture Loss Recovery and AI-Assisted Dryer-Cooler Optimization System

Deployment for reduction in moisture variation, prevention of sustained bad operation, and measurable improvement using QC lab data

Reducing Processing Losses and OverDrying is Key to Increase Profitability

Our Ai-Algorithms have Evolved to Reduce Processing Losses by 25-50%

Designed For Fluctuating Operating Conditions & Easy Adoption

Consistent Quality | Increased Productivity | Great Energy Saving | Meaningful Insights & Reports

1. Executive Summary

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Moisture shrinkage loss of 1–2% is presently normalized in feed manufacturing operations. For a plant producing 50,000 tons per year across four lines, this translates into approximately 500–1,000 tons of sellable product lost annually through evaporation. A significant part of this loss is not inherent to the process. It comes from sustained off-target operation, delayed correction, stale sensor calibration, and excessive dependence on operator response.

Optify's system is designed to first cap this downside before optimizing upside.

The first intervention is intentionally simple and measurable: install an online moisture sensor at the cooler outlet and use it as an early-warning and corrective-control layer. When high-moisture discharge is detected, the system triggers a bounded automatic response using the existing buffer volume available in the dryer and cooler. Line speed, heat, exhaust, and airflow are adjusted before the issue becomes sustained bad operation. This gives the process immediate first aid at the first instance of high-moisture discharge, especially during gate opening, SKU transition, or unstable running.

However, an online moisture sensor alone is not enough. If it is not continuously calibrated across SKUs, recipes, and operating conditions, it can create false alarms, stale readings, and incorrect process control. Optify addresses this through SKU-specific calibration models that continuously learn from online moisture data, QC lab results, and historical production behaviour. This reduces SOP burden on the plant team while keeping the sensor sharp enough for alarms, control decisions, and autonomous correction. Optify has filed a patent covering this calibration methodology.

Beyond moisture sensing, Optify integrates the key operating variables of the vertical dryer and cooler into one AI-assisted operating layer. This includes recirculation fan speed, exhaust fan speed, steam flow, airflow rate, air temperature, inlet and exhaust dew points, dryer and cooler levels, throughput changes, and QC lab results. The purpose is not to replace the existing automation system. The purpose is to sit above it, reduce operator cognitive load, prevent sustained bad operation, and improve consistency using the plant's own data.

The system enables operator-independent corrective action using multiple live data streams and online moisture sensing. It adjusts exhaust and recirculation speeds, adapts to changing ambient air conditions, monitors steam and airflow health, supports humidity sensor reliability, maintains moisture calibration, and applies interlocks linked to throughput changes. The result is a bounded, reversible optimization layer that works with existing plant automation instead of disrupting it.

To demonstrate effectiveness, Optify proposes one system as the first deployment. The pilot objective is a minimum 20% reduction in moisture standard deviation, measured through the customer's own QC lab data before and after Optify operation. The system is priced for approximately two-year payback at only 0.1% improvement in average retained moisture. Any gain beyond that remains entirely with the customer.

The proposed timeline is one month for delivery after advance release, followed by one month for commissioning. An additional one-month update period is reserved to review plant data, validate performance, refine calibration, and finalize the next phase of rollout.

The project is designed for low adoption friction. Improvements are visible in the customer's own QC data. The user interface is simplified for mill adoption. The system reduces avoidable
moisture loss without forcing the workforce to learn a complex new control platform. It also includes one-button hardwired fallback to legacy operation, allowing the plant to instantly return to its existing control mode if required.

Optify's approach is not a generic IT or digital transformation project. It is a surgical OT intervention for feedmill process control, focused on the specific operating points where mathematics can reduce dependence on human intuition. The system does not ask AI to rediscover process physics already known to the instrumentation and process teams. Instead, it uses process rules, historical plant data, local edge databases, and deterministic machine-learning blocks to identify where bounded automatic action can improve stability, reduce moisture variation, and prevent sustained off-target operation.

Optify's approach is conservative by design: first prevent avoidable loss, then stabilize moisture variation, then scale autonomous optimization across lines. This makes the first system a measurable pilot, a downside-containment tool, and a practical foundation for wider plant-level deployment.

Piyush Patil

Founder – Optify Industrial Solutions Private Limited

2. Present Operating Problem

A. Moisture Shrinkage Loss

Weight Shrinkage is Defined as difference between weight of raw materials vs weight of sellable goods. For a Shrimp feed plant with 4 lines, each producing 12,500 tons/ yr, the yearly weight loss is calculated below

Plant production	Normalized shrinkage	Sellable weight loss
50,000 TPA	1%	500 tons/year
50,000 TPA	2%	1,000 tons/year

Even partial recovery of this loss has a meaningful commercial impact.

B. Why This Loss Persists

i) Delayed feedback of moisture from laboratory
ii) Delayed operator correction after lab feedback
iii) Shifting Ambient air thermodynamics
iv) Delicate air flow balance in recirculating dryers

C. What challenges occur when automating them

i) Online Moisture Sensor Calibration Drifts
ii) Online moisture sensor calibration needs separate calibration for different SKU
iii) Exhaust humidity sensor gets clogged due to deposition of foreign matter

A. Online Moisture Sensing Layer

i) Online Moisture Sensor installed at cooler Or Sieve outlet
ii) Continuous comparison against target moisture
iii) High moisture alarm
iv) Data logging by SKU, recipe, batch, shift
v) Edge data analytics to reject outlier operations

B. Corrective Control Layer

i) Bounded automatic response during high moisture discharge
ii) Heat / Steam correction
iii) Exhaust fan correction
iv) Recirculation fan correction
v) Phase 2 – Mixer Water addition

The system does not make unrestricted control changes. It acts within defined process limits agreed during commissioning.

C. SKU-wise Calibration Layer

i) Online Moisture vs QC lab correction
ii) SKU specific calibration models
iii) False alarm reduction
iv) Lower operator SOP burden

Optify has filed a patent covering its continuous calibration methodology

D. Dryer – Cooler Digital Twin Layer Variables

i) Recirculation fan speed
ii) Exhaust fan speed
iii) Steam flow
iv) Airflow
v) Air Temperature
vi) Inlet and Exhaust dew points
vii) Dryer and cooler levels
viii) Throughput
ix) QC lab moisture
x) Online microwave moisture readings

These signals are used to identify operating zones, detect sustained off-target operation and recommend or execute bounded corrective action.

E. User Interface and Fallback Layer

i) Simple operator screen
ii) Alarm explanation
iii) Current operating mode
iv) Recommended action / automatic action
v) One button fallback
vi) Hardwired legacy bypass

4. Scope Of Supply

A. Hardware

i) Edge Server
ii) Signal Interface Panel
iii) Required IO integration
iv) Networking and communication interfaces
v) Fallback hardware
vi) Humidity sensor and Moisture sensor mounting accessories

B. Software

i) Data Acquisition
ii) SKU wise calibration model
iii) Alarm and corrective control engine
iv) Dryer/cooler operating model
v) Dashboard
vi) Reports
vii) Data logging
viii) Phase 2 – LLM based Diagnostics

C. Engineering Services

i) Site study
ii) IO mapping
iii) Sensor Installation support
iv) Commissioning of panel
v) Baseline data collection
vi) Calibration setup
vii) Operator training
viii) Performance Validation

D. Exclusions

i) Civil Works
ii) Cable tray work
iii) Mechanical Works
iv) PLC/DCS changes
v) Internet connectivity from router
vi) Online Moisture sensor (Directed Procurement)
vii) Humidity Sensors (Directed Procurement)
viii) Level, Temperature, flow meters, VFDs
ix) IT peripherials including monitors, TV, Keyboards and Mouse set
x) Mechanical Installation, electrical wiring and laying till panel

A. Phase 1 - Pre-commissioning Study

i) Process walkthrough
ii) Dryer/cooler mapping
iii) SKU list
iv) Moisture targets
v) QC lab method review
vi) Existing automation review
vii) IO availability check

B. Phase 2 - Installation

i) Panel interconnection
ii) Edge server installation
iii) Wiring and IO integration
iv) Signal validation
v) Dashboard setup

C. Phase 3 – Baseline Capture

i) Online moisture data
ii) QC lab data
iii) SKU wise trends
iv) Ambient data
v) Operator action history
vi) Bad-operation events

D. Phase 4 – Controlled Automation

i) Alarm validation
ii) Calibration model activation
iii) Bounded control activation
iv) Fallback testing
v) Operator training

E. Phase 5 – Performance validation

vi) Before/after QC comparison
vii) Moisture standard deviation reduction
viii) Average moisture analysis
ix) Downtime / bad operation event review
x) Next-line rollout recommendation

6. Performance Guarantee / Pilot Objective

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A. Primary Measurable objective

Minimum 20% reduction in moisture standard deviation, measured through customer QC lab data before and after Optify Operation.

B. Secondary Commercial upside

Improvement in average retained moisture will be calculated separately. At only 0.1% improvement in average moisture retention, the system is priced for approximately 2 year payback.

C. Validation Method

i) Baseline period
ii) Optify – controlled period
iii) Same SKU comparison when possible
iv) QC lab data as primary proof
v) Online sensor data as supporting proof
vi) Exclusion of abnormal production days if mutually agreed

7. Commercial Proposal

Item	Description	Actual
System Price	One-line deployment price	
Advance	35% of project cost + taxes	
Pre dispatch	35% of project cost + taxes	
Commissioning	20% of project cost + taxes	
Validation	10% of project cost + taxes	
Delivery	6 week after advance	
Commissioning	1 month after installation readiness	
Update / review period	1 additional month	
AMC period	1 year included	
Warranty	1 year included	
Additional line pricing	Additional lines deployed at location	

Upon successful validation of the first system, the same architecture can be replicated across remaining lines with lower engineering effort, since plant-level integration, calibration logic, and operating framework will already be established and this is reflected in additional line pricing.

8. Risk Containment and Plant safety

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A. Control Limits

i) System acts only within agreed process bounds
ii) No unrestricted actuator movement
iii) Manual override available
iv) Automatic action can be disabled

B. Fallback

i) One button hardwired fallback
ii) Legacy operation remains intact
iii) Plant can continue running without Optify

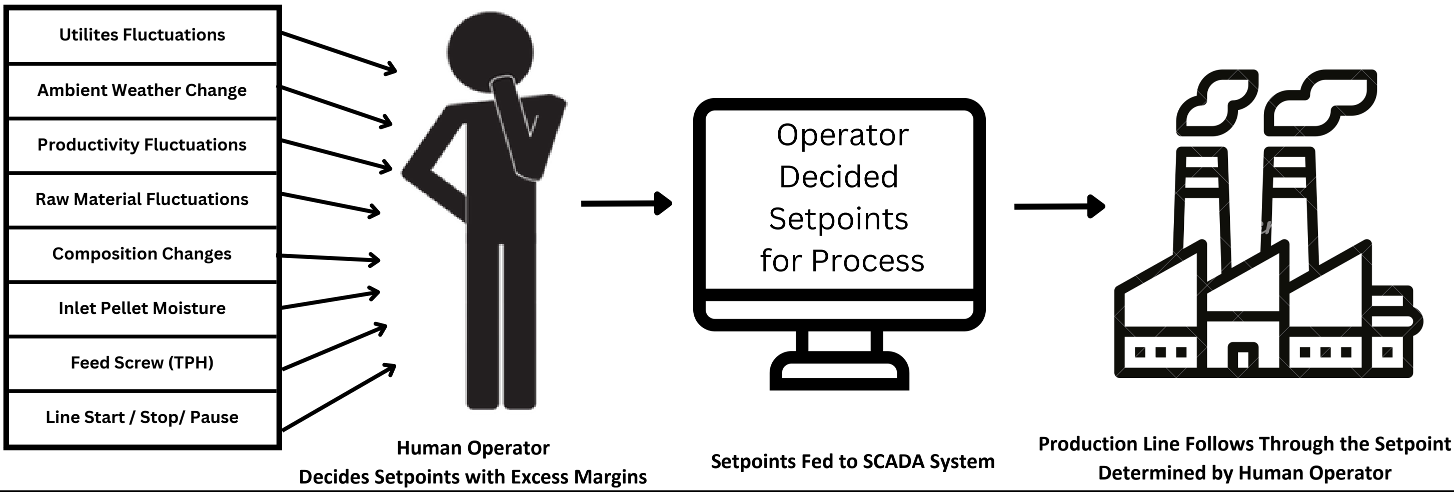
C. Data and Internet

Core control and optimization run locally at edge. Internet connectivity is used only for remote logging, monitoring, remote diagnostics, model review, updates and optional higher compute analysis. Production control does not depend on continuous internet availability. Yet customer to provide continuous internet connectivity to enable monitoring and logging layer.

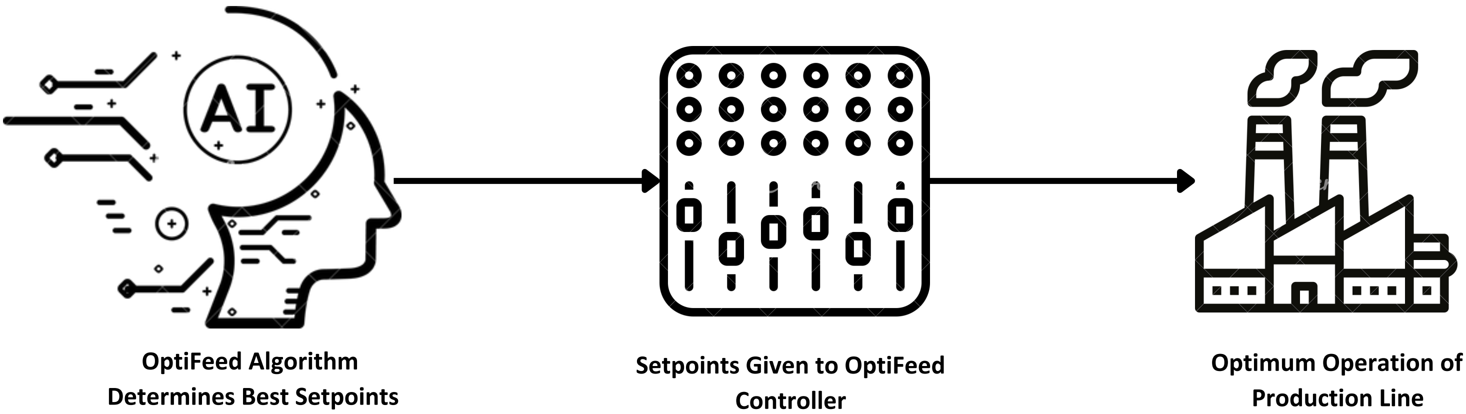
D. Responsibility boundary

i) Optify provides optimization layer
ii) Existing PLC/DCS remains primary automation layer
iii) Plant team retains final operational authority
iv) Safety interlocks remain with existing plant automation

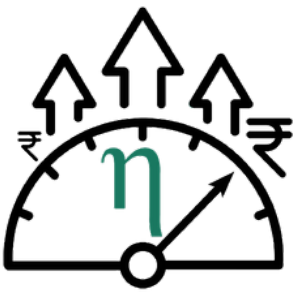
Limitations In Cognitive Capability Of Operator Results In Processing Losses



AI- ML Based Algorithm Produce More Sellable Material From Same Raw Material



Increased yield and reduced operator fatigue made possible by Ai-Technology



Higher profitability made possible by reducing processing losses

Autonomous data analytics and reports for tracking improvements in the production line

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Date: 20 April 2022
Rajahmundry, AP, India

Subject: **Optify Dryer + Cooler Optimization and Control System Performance**

Dear Sir,

This is to congratulate Optify team regarding the good performance of their shrimp feed dryer + cooler optimization and control system on Line 1 of CPF Rajahmundry plant.

The Optify system is fully autonomous and operates without requiring human judgement in normal operations.

In the **1-week demonstration period** from 24 March 2022 to 30 March 2022 (148 hourly lab samples), the following results were observed:

- **Average moisture - above 10.6%**
- **Standard deviation - below 0.4%.**
- **40% reduction in standard deviation** compared to the previous 1-month SCADA mode operation
- 3x reduction in the high moisture instances (above 11.1%)
- 5x reduction in moisture swings (>1% difference within 1 hour)
- 3x reduction in continuous high moisture instances (>11.1% for 2 hours)
- 100% avoidance of continuous very high moisture instances (>11.4% for 2 hours)

We are happy with the system's performance. We shall be planning for further expansion of more systems from Optify Industrial Solutions Pvt. Ltd.

We wish Optify team the best in their future endeavor.

Best regards

Mr. Thossaphorn Chattong
Department Manager
Production Improvement & Innovation Center - India
CPF (India) Private Limited

9. Next Steps

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A. Approval of first-line pilot

B. Site readiness and IO confirmation

C. Advance release

D. Delivery in 6 weeks

E. Commissioning and baseline validation

F. Performance review

G. Commercial rollout decision for remaining lines

The proposed first deployment allows the plant to validate Optify on one line with measurable QC data, limited operational risk, and a clear path to scale across all lines after performance review

Back Of The Brochure Benefits Calculation

Tons of finished goods my mill produce				(a) Tons/Annum
Price at which I sell the feed to my customers				(b) ₹/Ton
Moisture mean increase potential	0.25%	0.5%	1%	(c) Improvement
Extra sellable goods produced from same raw material (a) x (b) x (c)				₹/Annum

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